

Dairy Cattle Nutrition and the Environment

INTRODUCTION

Agriculture has seen a tremendous increase in productivity over the past century in the United States through the use of various technologies. Improved housing facilities, waste handling, breeding, and feeding balanced rations allowed livestock to be raised in increasingly larger and more concentrated animal feeding operations. The intensification in the U.S. dairy industry is such that farms with more than 500 milking cows accounted for 63 percent of the milk supply in 2012 (USDA, NASS, 2013), up from 45 percent a decade before (USDA, 2005). Consolidation of large numbers of dairy cattle into small land areas to improve the efficiency of milk production may contribute to environmental problems unless animals are fed and managed properly, including consideration of number of animals per unit of land available for manure application. From an environmental standpoint, the primary concerns are the nutrients nitrogen (N) and phosphorus (P); the greenhouse gases carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O); and other odorous compounds such as ammonia (NH₃), volatile organic compounds (VOCs), and hydrogen sulfide (H₂S) (NRC, 2003). Excretion of excess minerals in manure and their application in soil raises salinity and toxicity concerns, particularly in irrigated fields.

Dairy cattle play a key role in human food production by converting forages and poor-quality feeds into human edible products. However, this conversion is associated with an environmental cost, which can be unavoidable, that is, as a by-product of a necessary fermentation process or avoidable, for example, nutrients consumed in excess of requirement (Dijkstra et al., 2013a). Dairy cattle must be fed to meet their requirements with minimal excesses of nutrients in the diet if the efficiency of nutrient use and milk production by dairy cows are to be maximized and nutrient losses to the environment reduced. In 2008, the contribution of the entire dairy sector to the U.S. greenhouse gas (GHG) emissions was estimated to be 134 Tg CO₂ equivalents (CO₂e) or 1.9 percent of the U.S. total, with CH₄, N₂O, and CO, contributing 44, 13,

and 41 percent, respectively (Thoma et al., 2013). Although the dairy industry's environmental footprint is small compared to other industries (e.g., oil and gas), there has been continued pressure to reduce its footprint through national and state regulations. Reducing dairy production's impact on the environment such as air, soil, and water quality will contribute to the industry's long-term environmental sustainability (Kebreab, 2013; von Keyserlingk et al., 2013).

METHANE

CH₄, which has a global warming potential 28 times that of CO₂ over a 100-year horizon (IPCC, 2013), emitted from dairy operations is a significant contributor to GHG emissions. In 2014, dairy cattle contributed 25 percent of total enteric CH₄ emissions from livestock (USEPA, 2014). CH₄ emissions represent a loss of about 3.8 to 7.4 percent (5.6 percent on average) of gross energy intake (GEI) in U.S. dairy cattle (Kebreab et al., 2008a). Most CH₄ production occurs in the reticulorumen, with only 13 percent produced in the lower tract (Murray et al., 1976), and with rectal emissions accounting for about 2 to 3 percent of the total CH₄ emissions from the animal (Murray et al., 1976; Munoz et al., 2012).

Factors Affecting Methane Emissions

CH₄ production is positively and linearly related to the amount of feed consumed. Feed intake accounts for 60 to 80 percent of the variation in CH₄ production (Mills et al., 2003; Ellis et al., 2007; Moraes et al., 2014). The rest of the variation could be accounted for by differences in nutrient composition, uptake and utilization, and other factors. Although absolute CH₄ production (g/d) varies by breed, several studies have shown that breed has little impact on CH₄ produced per unit of intake (CH₄ yield) or per unit of product (CH₄ intensity) in ruminants (Fraser et al., 2014; Moraes et al., 2014). Ellis et al. (2010) evaluated enteric CH₄ production equations used in whole-farm models and concluded that

the simple, more generalized equations performed worse than those that attempted to represent important aspects of diet composition. The type of dietary carbohydrates (fiber versus nonfiber) fermented in the rumen plays a major role in determining the profile of rumen volatile fatty acid (FA) production (Murphy et al, 1982; Mills et al, 2001) and consequently CH_4 production (Moe and Tyrrell, 1979). Fermentation of fiber compared to starch results in greater acetate production, which increases hydrogen production and, consequently, enteric CH_4 production (Bannink et al, 2008). The same authors reported a shift toward production of propionate as pH in the rumen decreased, which uses hydrogen in the rumen and reduces enteric CH_4 production. Thermal stress may influence CH_4 production. Under prolonged cold conditions, Bernier et al (2012) reported CH_4 yield of 5.2 percent of GEI compared to 7.1 percent of GEI under thermoneutral conditions. This is related to increased rumen rate of passage, which ultimately reduces the extent and rate of rumen fermentation (see Chapter 5). Heat stress, on the other hand, may increase CH_4 yield due to longer retention of feed in the rumen or may decrease CH_4 yield because of lower feed intake and reduced rumen pH levels in response to reduced cation availability, as well as a consequent rise in the acetate to propionate ratio. Further research is required to quantify the effect of heat stress on enteric CH_4 emissions.

Mitigation Options to Reduce Enteric Methane Emissions

Hristov et al (2013a) extensively reviewed enteric CH_4 mitigation options and identified several opportunities. The mitigation strategies can be classified into (1) feed manipulation, (2) rumen modifiers, and (3) increasing animal production through genetics and management (Knapp et al, 2014). Feed manipulation through nitrate (NO_3) inclusion was considered to have the highest potential mitigation effect because up to 50 percent reduction in enteric CH_4 production in sheep and cattle has been observed due to NO_3 provision as an alternative electron acceptor (van Zijderveld et al, 2010, 2011; Hristov et al, 2013a). Several studies have shown that lipids have a suppressive effect on rumen microbes and CH_4 production (Martin et al, 2010; Grainger and Beauchemin, 2011). Eugene et al (2008) reported a 9 percent reduction in CH_4 production in dairy cows due to lipid supplementation, but dry matter intake (DMI) was reduced, so there was no difference in CH_4 yield (i.e., CH_4 production/kg DMI). Moraes et al (2014) quantified the response to dietary fat and, with all dietary factors being equal, observed an average decrease of 0.045 to 0.09 Meal (0.19 to 0.38 MJ) of CH_4 for every percentage increase in dietary fat. A meta-analysis by Knapp et al (2014) on the effect of lipid source showed that each percentage unit of diet crude fat (CF) from rumen-inert, seed, oil, and endogenous lipid sources decreased CH_4 intensity (i.e., CH_4 per unit of energy-corrected milk) by 0.7810, 0.20, 0.71±0.20, 1.12±0.20, and 1.01±0.38 g/kg, respectively. Greater DMI, feeding nonstructural carbohydrates, and improving forage

quality have low to medium potential impacts (Hristov et al, 2013a) and were expected to reduce enteric fermentation by 5 to 15 percent (Knapp et al, 2014). In view of competition with human edible feed resources, Hristov et al (2013a) recommended increasing forage digestibility and digestible forage intake among the major CH_4 mitigation practices.

Rumen modifiers considered to have low potential effects include ionophores and tannins (Hristov et al, 2013a). Apuhamy et al (2013) quantified the effect of an ionophore (monensin) in dairy cows to be -1216 g CH_4 /d (mean CH_4 production = 338 g/d) when adjusted for dose, DMI, and lipid intake. Care should be taken in extending CH_4 reduction results from in vitro studies to the commercial farm. For example, supplementing tea saponin (0.52 percent dry matter [DM]) reduced methanogenesis in vitro but increased CH_4 yield in vivo (Guyader et al, 2017). Until recently, the use of rumen modifiers or additives to reduce CH_4 production has been less successful compared to diet manipulation. However, several experiments have shown the potential for an inhibitor, 3-nitrooxypropanol, to reduce enteric CH_4 production in beef and dairy cattle (e.g., Hristov et al, 2015). Dijkstra et al (2018) conducted a meta-analysis using 11 published studies and reported that at an average dose of 81 mg/kg of DM, 3-nitrooxypropanol reduced CH_4 production 39 percent in dairy cattle. Bromoform and chloroform are halogens that have been found to interfere directly with the methanogenesis pathway (Goel et al, 2009). The red microalgae (*Asparagopsis* spp.) bromoform and other halogens, which have antimethanogenic properties, reduced enteric CH_4 emissions in vitro (Kinley et al, 2016) and in vivo (Roque et al, 2019). Up to 60 percent reduction in enteric CH_4 production has been observed at 1 percent of organic matter (OM) inclusion rate, but the authors caution that further work is needed to determine the long-term effects on productivity and animal health (Roque et al, 2019). Effects of these and other rumen modifiers on CH_4 production have been recently reviewed (Honan et al, 2021).

Management strategies to reduce CH_4 emissions were reviewed by Hristov et al (2013b). Increased productivity is considered as having greatest potential because of dilution of maintenance. From 1990 to 2012, enteric CH_4 emissions from dairy cattle increased 6 percent, cow numbers decreased 2 percent, and milk production increased 36 percent, indicating that while emissions per head increased, there was a 22 percent decline per unit of milk produced (USEPA, 2014). Capper et al (2009) estimated that the total carbon footprint for the entire dairy industry was reduced by 41 percent in 2007 compared to 1944 in the United States. In the Netherlands, from 1990 to 2008, yield of fat- and protein-corrected milk (FPCM) increased by 34 percent and CH_4 production increased by 16 percent per cow, but CH_4 intensity per unit FPCM decreased by 13 percent (Bannink et al, 2011). Based on modeling projections, precision diet formulation on a weekly and monthly basis may improve animal performance and consequently CH_4 emission intensity (White and Capper, 2014). Other management strategies with low to medium

potential to mitigate enteric CH₄ emissions include use of recombinant bovine somatotropin, growth promoters, genetic selection, improved animal health, reduced animal mortality, fibrolytic enzymes, and reduced forage maturity (Hristov et al., 2013b; Tewoldebrihan et al., 2017). Mitigation options that can be implemented in intensively managed dairy production systems currently (2020) may have a combined potential to reduce enteric CH₄ intensity by 15 to 30 percent (Knapp et al., 2014). However, using effective feed additives may reduce CH₄ intensity by more than 50 percent.

Enteric Methane Prediction Equations

Assessment of the efficiency of animal production and its subsequent environmental footprint requires quantification of each nutrient consumed, used, excreted, or lost to the environment. Mathematical models are widely used to predict the environmental impact of livestock operations and can be used to assess mitigation options and policy decisions. Appuhamy et al. (2016) evaluated 38 extant models developed to predict enteric CH₄ emissions from lactating dairy cows. The authors collected an extensive data set from around the world and evaluated each model with regional data. For North America, the highest-ranked model (which is also the recommended equation for dairy cows) was a modified version of a model developed by Nielsen et al. (2013) that uses DMI (kg/d), estimated digestible neutral detergent fiber (dNDF, percentage of DM), and FA (percentage of DM) contents:

$$\text{Methane (Mcal/d)} = 0.294 (+0.019) \times \text{DMI} - 0.347 (+0.093) \times \text{FA} + 0.0409 (+0.012) \times \text{dNDF} \quad (\text{Equation 14-1a})$$

Appuhamy et al. (2016) also evaluated previous and updated versions of the models (tier 2) recommended by the Intergovernmental Panel on Climate Change (IPCC, 1997, 2006). The earlier version of the IPCC model uses CH₄ conversion factor (Y_m, percent of GEI) of 6.0 percent, which is more in agreement with literature data from North America (5.7 ± 0.9; Appuhamy et al., 2016; Jayasundara et al., 2016) and also in agreement with Kebreab et al. (2008a), who reported 5.6 percent compared to the updated IPCC Y_m of 6.5 percent. Similarly, in their evaluation of CH₄ prediction equations used in whole-farm models, Ellis et al. (2010) reported better predictive capacity of the tier 2 method using Y_m of 6.0 percent compared with Y_m of 6.5 percent.

Moraes et al. (2014) developed equations to estimate CH₄ production for heifers and nonlactating dairy cattle based on indirect calorimetry measurements containing 414 and 591 records, respectively. The recommended equation for heifers is as follows;

$$\text{Methane (Mcal/d)} = -0.038 (0.071) + 0.051 (0.001) \text{ GEI} + 0.0091 (0.0014) \text{ NDF} \quad (\text{Equation 14-1b})$$

For nonlactating dairy cows, the recommended equation is the following:

$$\text{Methane (Mcal/d)} = 0.69 (0.048) + 0.053 (0.001) \text{ GEI} - 0.045 (0.012) \text{ CF} \quad (\text{Equation 14-1c})$$

In both equations, GEI is in Mcal/d, and NDF and CF are in percentages of DM.

Manure Methane and Volatile Solids Prediction Equations

According to IPCC (2006) tier 2 guidelines, CH₄ emissions from manure are determined based on estimated volatile solids and emission factors for various manure management systems. OM in livestock manure consisting of biodegradable and nonbiodegradable fractions are known as volatile solids. Using an extensive data set, Appuhamy et al. (2014) and Appuhamy et al. (2018) developed prediction equations to calculate volatile solids:

$$\text{Volatile solids (kg/d)} = 0.364 (\pm 0.007) \text{ DMI} + 0.026 (\pm 0.004) \text{ NDF} - 0.078 (\pm 0.008) \text{ CP} \quad (\text{Equation 14-2})$$

where DMI is in kg/d, and crude protein (CP) and NDF contents are in percentages of DM.

However, only digestible OM generates CH₄; therefore, CH₄ emissions should be based on digestible volatile solids (i.e., volatile solids—lignin). Appuhamy et al. (2018) developed a mathematical model for estimating digestible volatile solids outputs by lactating dairy cows:

$$\text{Digestible volatile solids (kg/d)} = 0.334 (\pm 0.007) \text{ DMI} + 0.029 (\pm 0.006) \text{ HC} - 0.058 (\pm 0.008) \text{ CP} \quad (\text{Equation 14-3})$$

where DMI is in kg/d, and CP and hemicellulose (HC = neutral detergent fiber—acid detergent fiber contents) are in percentages of DM.

It is recommended to use the volatile solids prediction equations given above when predicting CH₄ from manure management using the IPCC (2006) methodology.

NITROGEN

N is of primary environmental concern because of losses of organic N and ammonium via wind and water erosion, NH₃ through volatilization, NO_x through leaching and denitrification, and oxides of N as a result of nitrification-denitrification processes (Eckard et al., 2010; Cavigelli et al., 2012). The main causes of N loss from the animal are inefficient utilization of feed N in the rumen, undigested feed and microbial true protein, microbial nucleic acids synthesized in the rumen, N use inefficiency for maintenance and milk protein synthesis,

and amino acids (AAs) absorbed in excess of requirement (Dijkstra et al., 2013c; Hristov et al., 2013b). The impact of these N losses from agricultural systems on surface- and groundwater quality, soil pH, biodiversity, and pathogens has been well documented (Tamminga, 1992).

Factors Affecting Nitrogen Excretion

The main driver of N losses from cattle is N consumed in feed. Dairy cows secrete in milk, on average, 21 to 33 percent of the N they consume (Calsamiglia et al., 2010), with almost all of the remaining N excreted in feces and urine. Using a large database, Reed et al. (2015) calculated an average total manure N excretion of 69 percent of N intake. In a meta-analysis, Huhtanen and Hristov (2009) concluded that dietary CP concentration is the most important dietary factor influencing milk N efficiency, with ruminal degradation of CP being of lesser importance. Similarly, of all single dietary and animal factors evaluated in a meta-analysis by Spek et al. (2013c) to predict N excretion in urine, dietary CP concentration and milk urea N level were by far the best predictors. Differences in amount and, to a smaller extent, digestibility of N in feed affect not only the total amount excreted but also the partitioning of N into milk, urine, and feces (Castillo et al., 2001b; Kebreab et al., 2002). Such a distinction of manure N excretion into fecal and urinary N excretion is of significance, because variation in dietary N supply will affect urinary N output, which is more susceptible to leaching and volatile losses than fecal N and of larger importance to reduce environmental impact (Dijkstra et al., 2013b). Dijkstra et al. (2013c) calculated the theoretical upper limit of N use efficiency to be 43 percent at maximal milk secretion for a cow weighing 650 kg and producing 40 kg/d of fat- and protein-corrected milk.

Mitigation of Nitrogen Losses from Cattle Operations

The wide variation in efficiency of conversion of feed N into products suggests that major improvements in reducing N excretion are possible (Dijkstra et al., 2013c). There is little opportunity to reduce N losses related to incomplete digestion of microbial protein, synthesis of microbial nucleic acids, and animal maintenance requirements (Dijkstra et al., 2013c). The most effective strategy to reduce N excretion and limit impact on the environment is to decrease the dietary CP content (Castillo et al., 2000; Kebreab et al., 2001; Spek et al., 2013c). Several studies have shown that total N excretion increases as dietary N intake increases, with urinary N excretion increasing at a greater rate than fecal excretion (Castillo et al., 2000; Huhtanen and Hristov, 2009; Kebreab et al., 2010). Feeding a diet that contains CP above the requirement will also increase energy expenditure slightly to cover the cost of urea synthesis. Because CP is usually a costly nutrient, feeding excess CP also inflates feed costs to producers. Reduction of N intake decreases the amount

of N excreted (Huhtanen et al., 2008) and may improve N use efficiency (Kebreab et al., 2010). Furthermore, as milk production per cow increases, because of dilution of maintenance N use, efficiency should increase with a concomitant reduction in N excretion per unit of product (Capper, 2011). However, reducing dietary N below requirement will impair productivity (Law et al., 2009). In addition to reduction of CP content, other dietary strategies such as optimizing rumen fermentation and microbial protein synthesis, passage of nutrients to the small intestine, and efficiency of absorbed amino acid (AA) utilization for milk protein synthesis can all be effective mitigation options (NRC, 2001). Efficiency of N utilization is affected by availability of energy, and generally milk production is increased with greater concentration of energy in the diet. In a multivariate analysis, Reed et al. (2014) showed that as the metabolizable energy content of the diet increases, efficiency of N use increased with diminishing returns. Modifying rumen microflora, particularly those involved in peptide degradation and AA deamination, may increase efficiency of N utilization (Calsamiglia et al., 2010). Further work is needed to better understand factors controlling urea transport across the rumen wall and take advantage of ruminants' ability to recycle urea (Calsamiglia et al., 2010). However, when N intake is low and rumen microorganisms might benefit from additional supply of N, close to 100 percent of urea synthesized in the liver is recycled to the gastrointestinal tract (Reynolds and Kristensen, 2008), and consequently, there is little potential for urea recycling to compensate low CP diets. Postrumen metabolism of AAs in the portal-drained viscera and the liver contributes to N excretion. Lapierre et al. (2005) estimated, on average, 35 percent of AAs are lost during absorption, and the liver removes 45 percent of absorbed AAs, giving rise to significant amounts of urea excreted in urine. Further opportunities to decrease N losses postrumen also arise from proper balancing of diets for individual AAs (Haque et al., 2011). Feeding diets that contain lower concentrations of CP supplemented with balanced quantities of rumen-protected AAs should reduce excretion of N in urine. Milk protein yield is particularly responsive to essential AA supply to the mammary gland, and improvements in the ability to model AA supply and use should improve N use efficiency.

Nitrogen Excretion Prediction Equations

Precise estimates of N excretion from livestock lead to better quantification of manure N, which is a basis for estimating N volatilization, leaching, runoff, and emission. Several models have been developed to predict N excretion from lactating dairy cattle and heifers to assess the efficiency of cattle production and calculate national inventories of N₂O emissions. Although the accompanying model will calculate N excretion based on the mass-balance approach, some equations are provided here to aid input-output type of analysis. For lactating cows, if DMI, dietary CP concentration, milk

yield, and milk protein concentrations are known, mass balance can be used to estimate manure:

$$\text{Manure N (g/d)} = (\text{DMI} \times \text{DietCP}) / 0.625 - (\text{Milk} \times \text{MilkCP}) / 0.638 - 5 \quad (\text{Equation 14-4})$$

where DMI and Milk are in kg/d, and DietCP and MilkCP are in percentages. The 5 is an estimate of body growth over three lactations (assumed to be 150 kg of growth over 915 days).

Johnson et al. (2016) evaluated 45 models to predict N excretion when milk protein yield is not known. The following equations were recommended for lactating dairy cows:

$$\text{Urine N (g/d)} = 12.0 (\pm 5.80) + 0.333 (\pm 0.011) \text{ N intake (g/d)} \quad (\text{Equation 14-5})$$

$$\text{Fecal N (g/d)} = -18.5 (\pm 3.59) + 10.1 (\pm 0.169) \text{ DMI (kg/d)} \quad (\text{Equation 14-6})$$

$$\text{Total Manure N (g/d)} = 20.3 (\pm 4.72) + 0.654 (\pm 0.009) \text{ N intake (g/d)} \quad (\text{Equation 14-7})$$

$$\text{Milk N (g/d)} = -19.0 (\pm 3.21) + 8.13 (\pm 0.245) \text{ DMI (kg/d)} \quad (\text{Equation 14-8})$$

For heifers and nonlactating cows, the following equations were recommended:

$$\text{Urine N (g/d)} = 14.3 (\pm 3.18) + 0.51 (\pm 0.12) \text{ N Intake (g/d)} \quad (\text{Equation 14-9})$$

$$\text{Feces N (g/d)} = 0.35 (\pm 1.73) + 0.32 (\pm 0.0064) \text{ N Intake (g/d)} \quad (\text{Equation 14-10})$$

$$\text{Total Manure N (g/d)} = 15.1 (\pm 2.50) + 0.83 (\pm 0.018) \text{ N Intake (g/d)} \quad (\text{Equation 14-11})$$

The root mean square prediction error (as a percentage of mean observed values), which measures the prediction performance, was 25, 16, 11, 14, 36, 17, and 13 percent for Equations 14-5 to 14-11, respectively.

Ammonia

Dairy production contributes to NH₃ emission, which can create human respiratory problems by forming fine particulate

matter with other compounds and animal health hazards when concentrations reach critical levels in confined spaces. NH₃ emission can also cause regional degradation of terrestrial and aquatic ecosystems (NRC, 2003; Kampa and Castanas, 2008) and represents a net loss of manure fertilizer value. NH₃ emission is regulated by the U.S. Environmental Protection Agency (EPA) through the Clean Air Act Amendments of 1990 (Pub. L. 101-549). About 90 percent of NH₃-N originates from urine N and the rest from feces within a 10-day period after excretion (Lee et al., 2011). Large variation in NH₃ emission estimates is reported mainly due to several factors affecting measurement, including wind, temperature, time of day, and year. In a recent meta-analysis, Bougouin et al. (2016) identified housing system, season, diet composition, and milk production as major factors affecting NH₃ emissions from dairy cattle. Hristov et al. (2011) calculated a daily average of 59 g of NH₃ emission per cow based on a compilation of studies, with a large standard deviation of 65 g/d. However, when averaged over a year, NH₃ emissions measured in open-lot dairy housing systems in Idaho, Texas, and California were more consistent and ranged between 120 and 150 g/d per cow (Leytem et al., 2011). Emissions from freestall and open-freestall dairies were lower at 10 to 100 g/d per cow (Leytem et al., 2013). In general, stall NH₃ emissions are much higher than pasture NH₃ emissions (McQuilling and Adams, 2015). On average for grazing cows, 25 percent of the N excreted in manure (Hristov et al., 2011) and 3 to 15 percent of urine N in the field or 4 to 52 percent from urine patches (Oenema et al., 2008) may be lost as NH₃ depending on soil type, moisture, temperature, wind speed, and the concentration and forms of N in urine. NH₃ emission can be reduced by dietary manipulation, increasing milk yield, manure treatment, and capture and treatment of emitted gases (Hristov et al., 2011; Bougouin et al., 2016). Decreasing the dietary CP content is probably the most effective strategy to decrease NH₃ emission from dairy manure (Bougouin et al., 2016) due to reduced N substrate in the excreta, particularly urine N, and as a consequence reduces the environmental impact (Frank et al., 2002; Frank and Swenson, 2002; Agle et al., 2010). Urine N, because it is 52 to 94 percent urea, is more susceptible to leaching and volatile losses than fecal N (Reynal and Broderick, 2005; Dijkstra et al., 2013b). Thus, reducing urinary N excretion will greatly reduce environmental impact. However, a trade-off with other losses of N may occur. If a smaller fraction of manure N is emitted as NH₃, losses of NO₃ or N₂O may increase (van Groenigen et al., 2008), and simultaneously, lowering all of these losses is a challenge.

Milk urea N has been used as a proxy for urine urea N because the two are correlated (Burgos et al., 2007; Powell et al., 2011, 2014). A number of factors, including body weight, urine production, and time and frequency of feeding and milking, have been shown to affect the relationship between milk urea N and urinary N excretion (Spek et al., 2013b). Powell et al. (2014) calculated that each 1-mg/dL

decrease in milk urea N was associated with a reduction in NH_3 emissions from manure of 7 to 12 percent. However, the relationship between milk and urine urea N is affected by mineral concentration of the diet, which negatively affected milk urea N but not urine urea N (Spek et al., 2012, 2013a; Eriksson and Rustas, 2014). Variation in urine volume also affected the relationship between milk urea and urinary N excretion. Care should therefore be taken when using milk urea N as a proxy for urine urea N (Spek et al., 2013b).

Nitrous Oxide

Nitrogenous excretions from dairy cattle can directly contribute to overall N_2O emissions (Hristov et al., 2011), which have a global warming potential 265 times that of CO_2 over a 100-year horizon (IPCC, 2013). In addition, indirect emissions of N_2O occur after atmospheric deposition of NH_3 and nitrogen oxides (NO_x) from animal housing and manure storage and various N sources from leaching and runoff (IPCC, 2006). In 2012, about 4 percent of N_2O emissions in the United States were attributed to the breakdown of N in livestock manure (USEPA, 2014). For N_2O to occur, manure must undergo several transformations mediated by microbes: hydrolysis and mineralization of organically bound N into ammonium (NH_4^+), nitrification to nitrite (NO_2^-), and NO_3^- in the aerobic environment followed by anaerobic reduction to elemental N, with intermediate production of N_2O and nitric oxide (NO) through denitrification (Li et al., 2012). Factors affecting N_2O emissions include temperature, moisture content, availability of easily degradable organic carbon, and oxidation status of the environment (Montes et al., 2013). Because of environmental conditions, the fraction of manure N lost as N_2O is generally below 2 to 3 percent, with a few studies reporting up to 10 percent (Groenigen et al., 2005; Luo et al., 2010). In open-lot dairies, Leytem et al. (2011) measured N_2O emissions ranging from 19 to 33 g/d per cow over a year, which were greater than for open-freestall dairies measuring 5 to 37 g/d per cow (Leytem et al., 2013). Urine is the main source of volatile N emissions; therefore, manipulating the route of N excretion is an important N_2O mitigation tool. The portion of urine N released as N_2O depends on the urinary N composition, soil type, soil wetness, and soil temperature. Emissions are relatively low when the soil is dry or very wet and relatively high when the water-filled pore space in soil ranges from 60 to 80 percent (Dijkstra et al., 2013b). In soil, urinary hippuric acid, creatine, and creatinine decompose more slowly than urea, and hippuric acid may act as a natural inhibitor of N_2O emissions (Dijkstra et al., 2013b). About 92 percent of N_2O emissions occur in cropping systems, and mitigation options related to such systems are reviewed by Cavigelli et al. (2012). Successful mitigation options in cropping systems such as adding urease and artificial nitrification inhibitors (e.g., dicyandiamide) are generally effective in controlling N_2O emissions (Clough

et al., 2009). Manure storage and treatment-based mitigation options for reducing N_2O emissions have been extensively reviewed by Montes et al. (2013). Prediction of N_2O emissions is challenging because of the various factors involved that have an impact on N_2O formation. Most of the quantification methods of N_2O emissions from the dairy sector are based on static emission factors from IPCC (2006), which do not account for the wide range of variability inherent in the dairy industry. As a result, the methodology is associated with a high degree of uncertainty and will not be appropriate to use for assessment of mitigation options. Additional research on quantification and impact of dairy production on N_2O and other GHG emissions in the United States is warranted.

Water Quality

Dairy operations tend to be concentrated in certain regions of the country, which results in long-distance redistributions of substantial amounts of N and other nutrients (Kellogg et al., 2000). Nutrients accumulate near dairy operations, often in quantities that exceed the nutrient needs of the crops being grown within reasonable transportation distance (Rosenstock et al., 2014). These nutrients contribute to hypereutrophication of estuaries and N leaching to ground-water (Kellogg et al., 2000). Contamination of groundwater can have negative consequences on water supplies and their suitability for human consumption and use (Townsend et al., 2003). For example, excessive consumption of NO_3^- in drinking water has been associated with methemoglobinemia or “blue baby syndrome” in humans, stomach cancer, and NO_3^- poisoning in animals (Pasten-Zapata et al., 2014). It also contributes to surface-water contamination over longer time frames as groundwater reenters surface drainage networks (Meals et al., 2010; Wick et al., 2012). NO_3^- export into adjacent surface water bodies may induce an increased level of nutrients (eutrophication) affecting biodiversity, mammals, birds, and fish adversely by producing toxins and reducing oxygen levels (Pasten-Zapata et al., 2014). Harter et al. (2002) assessed NO_3^- and salt leaching to shallow groundwater near dairies in San Joaquin Valley, California. They estimated minimum average annual groundwater NO_3^- -N and salt loading from manure-treated forage fields to be 280 and 4,300 kg/ha, respectively. Leaching rates for ponds were estimated to be about 0.8 m/year, at least locally. Over six decades, there was over a 10-fold increase in NO_3^- loading in two agricultural areas in California. Meeting safe drinking water standards would require leaching reductions of over 70 percent from current levels through reductions in excess manure application, which accounts for nearly half of all groundwater N loading, and through synthetic N management improvements (Rosenstock et al., 2014). In manure-treated fields, proper nutrient management will be a key to protecting groundwater quality, particularly in regions overlying alluvial aquifers (Harter et al., 2002).

VOLATILE ORGANIC COMPOUNDS

VOCs are defined as any carbonaceous compounds that participate in atmospheric photochemical reactions, excluding CO₂, carbon monoxide, carbonic acids, metallic carbides, or carbonates (USEPA, 2011). Some of the common VOCs include alcohols, aldehydes, ketones, esters, ethers, aromatic hydrocarbons, and halogenated hydrocarbons (Filipy et al., 2006). These gases are important air pollutants because they contribute to the formation of ozone, which is a constituent of photochemical smog in the presence of oxygen and sunlight (Carter, 1994). Some of the common VOCs emitted from dairy production include acetic acid from fermented feeds and manure (Shaw et al., 2007; Alanis et al., 2008); acetaldehyde from fermented feeds (Howard et al., 2010); ethanol from fermented feeds, manure lagoons, and housing (Filipy et al., 2006; Chung et al., 2010; Howard et al., 2010); methanol from enteric fermentation and manure (Shaw et al., 2007); and acetone from manure lagoons and housing (Filipy et al., 2006; Chung et al., 2010).

Good silage-making practices such as rapid filling, adequate packing, and the use of inoculants and preservatives can reduce VOC emissions from fermented feeds (Muck, 1988; Place and Mitloehner, 2013). Minimizing the exposed surface area of silos reduces emissions substantially because greater emissions occur within the first 12 hours of silage being exposed to air (Hafner et al., 2010). Water-soluble VOCs from animal housing could be reduced by flushing the barn floors with water (Chung et al., 2010); however, this may increase VOC concentrations in the lagoon (Place and Mitloehner, 2013). Biofiltration systems also have the potential to reduce VOC emissions by passing exhaust air from housing or manure storage systems through a filter containing micro-organisms and “trapping” emissions (Martens et al., 2001; Pagans et al., 2007). More research is required to understand and quantify the sources and the factors influencing the VOC emissions in dairy operations.

INTEGRATED APPROACHES

The ultimate objective in reducing the environmental burden of dairy production should be focused on net reduction because mitigation options that have one environmental benefit may negatively impact another. In this respect, the level of analysis (at animal, farm, or food chain levels) is of large significance. For example, van Middelaar et al. (2013) evaluated the effect of replacing grass silage with corn silage on GHG emissions at all three levels using a linear programming model in combination with a mechanistic model of enteric fermentation and life cycle assessment. Although at animal, farm, and food chain levels, this strategy reduced annual GHG emissions by 12.8, 17.8, and 20.9 kg CO₂e per ton FPCM, converting grassland into corn land resulted in nonrecurrent emissions of more than 900 kg CO₂e per ton FPCM. Although enteric

CH₄ emissions contribute the greatest to whole-farm emissions (Thoma et al., 2013), manure management, particularly N₂O emissions, is also a significant contributor, mainly due to greater global warming potential. The processes involved in CH₄ and N₂O production are often antagonistic because the former is produced under anaerobic conditions, whereas the latter requires oxygen; therefore, some practices such as composting that result in the reduction of CH₄ production may increase subsequent N₂O emissions (Montes et al., 2013). Ellis et al. (2012) showed that mitigation options aimed at reducing urinary N excretion may result in elevated CH₄ emissions depending largely on the type of carbohydrate consumed. CH₄ production declines if starch or digestible nutrients escaping rumen fermentation replace protein in the diet but rises if dietary fiber levels increase. Reducing dietary CP by increasing fiber will likely increase CH₄ production while decreasing N₂O. Dijkstra et al. (2011) estimated an increase of on average 0.30 g CH₄ per gram urinary N decrease for various nutritional interventions with grass silage-based diets aimed to improve milk N efficiency. Using standard emission factors for direct and indirect N₂O emissions, the estimated N₂O emission reduction (in CO₂ equivalents) resulting from decreased manure N output was more than offset by a rise in enteric CH₄ production. Similarly, Sauvant et al. (2014) reported that CH₄ production per kilogram digested OM decreased in a linear fashion with increasing dietary CP concentration, which will likely increase N₂O emissions. Analysis of 1,111 records from calorimetry chambers at USDA Beltsville, Maryland, using Holstein and Jersey cows also showed a weak negative relationship, possibly because degradation of OM may be enhanced by greater dietary CP concentration (Colmenero and Broderick, 2006; Nousiainen et al., 2009). As expected, CH₄ production showed a positive linear relationship with neutral detergent fiber (NDF) content and negative linear relationship with metabolizable energy intake. Animals fed low CP diets produced manure with a slower mineralization rate of N, which reduced N₂O emissions when land applied (Powell and Broderick, 2011); however, CH₄ emissions may increase because of higher OM content. If the amount of N is not sufficient for crop growth, the need for higher rate of inorganic N fertilizer inclusion might also increase overall N₂O emissions. Moraes et al. (2012) developed a linear programming model to formulate minimum-cost diets when environmental policies are present. In their evaluations, imposing CH₄ restrictions increased N losses from the animal. Van Middelaar et al. (2014) reported extruded linseed or NO₃ supplementation to reduce CH₄ intensity by 42 and 33 kg/ton FPCM at the animal level, respectively, but total GHG emission at the food chain level was reduced by only 9 and 32 kg/ton FPCM, respectively. In view of these results, a major challenge in reducing N losses from dairy cattle is to find an optimal nutritional balance without increasing enteric CH₄ production. The relationship between manure NH₃, volatilization and N₂O emission is also complex because if a mitigation technol-

ogy reduces NH_3 losses, the preserved ammonium may later increase soil N_2O emissions (Petersen and Sommer, 2011). On the other hand, losses of N as NH_3 reduce the availability of N for nitrification and denitrification processes and, consequently, N_2O formation. In their review, Montes et al. (2013) listed several opportunities for mitigation; however, only decreasing manure storage time was recommended for reducing both GHG and NH_3 emissions. In pasture-based production systems, improving forage quality is often accomplished by increasing N fertilizer application rates, which can have a negative impact on urinary N excretion and thus NH_3 and N_2O emissions.

MINERALS

Phosphorus

P is an essential mineral for ruminants (see Chapter 7). In dairy cattle, more than 55 percent of the P consumed may be excreted in manure depending on P availability, efficiency of feed conversion, and the amount of P consumed in excess of the animal's requirement (Kebreab et al., 2005b; Vitti and Kebreab, 2010; Klop et al., 2013). The rest is excreted mostly in feces with less than 1 g P/d in urine unless cows are fed 20 to 30 percent in excess of requirement (Wu et al., 2000). Apart from P being a finite resource, manure P has a potential to contribute to environmental degradation, particularly degradation of water quality. Manure phosphate applied to land is usually adsorbed onto soil particles, so it does not leach into water tables or waterways but builds up in the soil (Pierzynski et al., 1994). This becomes a problem when the P-laden soil particles are washed into surface water. In the United States, the median contribution of P from animal agricultural sources to the nation's watersheds is 26 percent, compared to 17 percent from commercial fertilizer and 3 percent for point sources (Smith and Alexander, 2000). Due to environmental impact, there have been substantial public interest and policies to reduce P excretion and manure P applied to agricultural soils. For example, the U.S. EPA is mandated to enforce P reductions of at least 0.6 million kg/year in the Chesapeake Bay watershed to maintain water quality (USEPA, 2009). There are several ways to reduce P excretion from dairy cattle. Kebreab et al. (2012) broadly divided mitigation options for P excretion into two categories: (1) increasing efficiency of P utilization and (2) improving or optimizing P availability in feed. Several surveys revealed that dairy cattle in the United States (Dou et al., 2003; Castillo et al., 2013) and Canada (Kebreab et al., 2008b) were routinely fed diets that include 0.45 to 0.50 percent P, which is in excess of animal requirements (NRC, 2001; Wu et al., 2001; Valk et al., 2002). Castillo et al. (2013) reported that the median P concentration in a typical California dairy diet was 1.3 times the NRC (2001) requirements. Excess P is excreted mainly in feces, and there is a positive linear relationship between P intake and P in excreta

(Kebreab et al., 2005b; Klop et al., 2013). Using a mechanistic model, Hill et al. (2008) predicted that total P in the diet had a greater effect on P excretion than any other diet fraction. In a meta-analysis of dairy cattle data, Klop et al. (2013) found P intake to be the major predictor of P excreted in feces, and small but significant effects of dietary NDF content, which increased fecal P excretion, and of dietary CP content and milk production level, which decreased fecal P excretion, were also found. Therefore, reducing P intake by matching animal P requirement with available P in the diet is the most effective way of mitigating P excretion in dairy cattle. Recent advances in biotechnology, such as nutritional genomics, may offer genome-tailored individual animal requirements to closely match up with dietary supply (Kebreab et al., 2012).

Availability of dietary P can be increased when nonruminants are fed exogenous phytase; however, because rumen bacteria synthesize phytase, exogenous phytase does not greatly affect P availability to ruminants (see Chapter 7 for additional discussion). Continued development of phytase through improved understanding of its ability to break down organic P may produce more effective classes of phytases, which could increase P availability, which should reduce the need for dietary P and decrease fecal P (Kebreab et al., 2013).

The amount and type of carbohydrate in diet can affect P use efficiency. Kebreab et al. (2005a) reported that dairy cows excreted up to 15 percent less P when fed readily available carbohydrate sources such as starch compared to structural carbohydrate sources. Using a greater amount of starch rather than fiber may have led to greater incorporation of nutrients by microbes, including P. Hill et al. (2008) showed that the efficiency of using P increased when the energy content of the diet increased due to greater P incorporation in milk. In animals on pasture, rotational grazing has reduced total P load in runoff by 64 percent and improved soil infiltration compared to traditional grazing (Haan et al., 2006). A survey of the northeastern United States showed that 13 percent of dairy producers use this method and had better economic and environmental performance than producers using traditional grazing systems (Winsten et al., 2010).

Phosphorus Excretion Prediction Equations

Alvarez-Fuentes et al. (2016) evaluated 10 extant models and developed empirical models to predict P output in feces. The authors developed two sets of models, using all variables that are correlated with P output in one set and, in the second, selecting variables that are routinely available in commercial dairies. Among the extant models, those developed by Weiss and Wyatt (2004) and Klop et al. (2013) performed best when evaluated with data collected from published studies conducted after 2000 (Equations 14-12 and 14-13, respectively).

$$\text{Fecal P (g/d)} = -2.3 (\pm 4.2) + 0.63 (\pm 0.046) \times \text{P intake (g/d)} \quad (\text{Equation 14-12})$$

$$\text{Fecal P (g/d)} = 19.9 (\pm 5.07) + 0.79 (\pm 0.060) \times \text{P intake (g/d)} - 1.04 (\pm 0.127) \times \text{Milk yield (kg/d)}$$

(Equation 14-13)

Among the new models developed by Alvarez-Fuentes et al. (2016), the following model had the best performance:

$$\text{Fecal P (g/d)} = 0.73 (\pm 0.03) \times \text{P intake (g/d)} - 0.37 (\pm 0.08) \times \text{Milk yield (kg/d)}$$

(Equation 14-14)

Comparing the three models (Equations 14-12 to 14-14), the more complex models that use P intake and milk yield had similar performances, which was slightly better than the simplest model requiring only P intake. Therefore, depending on availability of data, any of the three models are recommended for use.

Other Minerals

The limited land base of intensive dairy production necessitates importation of nutrients, including minerals in feed. If imported minerals do not leave the farm, mineral accumulation occurs, which has a negative impact on soil and water quality. Surveys have shown substantial mineral surpluses in dairy operations (Hristov et al., 2007; Castillo et al., 2013). In general, most dairy diets provide more mineral than NRC (2001) requirements. A survey of 39 commercial dairies in California showed that the median concentrations of minerals in diets above NRC (2001) requirements were 1.4x for calcium (Ca), 10.6x for magnesium (Mg), 10.5x for sodium (Na), 13.6x for chlorine (Cl), 1.6x for potassium (K), 1.4x for sulfur (S), 1.8x for copper (Cu), 24.7x for iron, 42.0x for manganese, 1.5x for selenium (Se), and 1.6x for zinc (Zn; Castillo et al., 2013). Depending on the source, drinking water can also add to total mineral consumption (Castillo et al., 2013), which, when excreted and applied to soil, can contribute to salinity problems, particularly in irrigated fields. The efficiency of utilization and excretion of minerals varies widely depending on the amount, requirement of the animal, the form of mineral (bioavailability), and interaction or coavailability between the minerals consumed. Excretion of minerals by cows fed various practical diets range 68 to 85 percent of intake for Ca, 86 to 93 percent for K, 95 to 98 percent for Mg, 45 to 95 percent for Na, and 58 to 87 percent for Cl (Meyer and Robinson, 2007). Taylor et al. (2018) developed models to predict Ca, Mg, and Se. The authors reported that DMI and mineral concentrations are strong predictors of mineral excretion.

Minerals such as Cu and Zn are less mobile in soil and may accumulate over time (Brock et al., 2006) and can be toxic to plants and foraging animals (Ferket et al., 2002). High Mg concentration in manure has been reported to inhibit crystallization of stable phosphate forms in sandy

soils, enhancing the release of P to the environment (Josani et al., 2005). Manure S can be a significant source of S emissions and odor because it is a precursor of hydrogen sulfide (NRC, 2003). Dietary manipulation to reduce mineral consumption and more precise formulation of rations to meet animal mineral requirements are the most likely mitigation strategies to reduce mineral excretions from livestock operations (Hristov et al., 2007).

TOTAL MANURE

Several equations have been developed to estimate urine and total manure (urine plus wet feces) output by dairy cows (Wilkerson et al., 1997; Bannink et al., 1999; Wattiaux and Karg, 2004; Nennich et al., 2005; van der Stelt et al., 2008; Weiss et al., 2009). Manure is mostly water with DM concentrations usually less than 15 percent. The major factors influencing output of manure are DMI, diet digestibility, and intake of certain minerals, most important, Na and K. N (CP) intake also has some effect. Intake of indigestible DM, rather than DMI, should be a better predictor of manure output. However, equations are usually developed for practical application, and in the field, diet indigestibility cannot be measured. Na and K are major drivers of water flux (see Chapter 7), and increasing intake of those minerals usually increases intake of water, which increases urine output and the water concentration of feces. Most of the published equations were derived from lactating cow data; however, Nennich et al. (2005) includes equations for heifers and dry cows that were derived from limited data. The equations used in the software were chosen because they included independent variables that are major sources of variation in manure output and readily available in most situations. Because of numerous other variables influencing manure output, these equations are not adequate for regulatory purposes. The following equations are used in the software to estimate manure output in kg/d for various classes of animals:

$$\text{Growing heifers: Manure output} = 4.16 \times \text{DMI}$$

$$- \text{BW} \times 0.0246 \quad (\text{Equation 14-15a})$$

(Nennich et al., 2005)

$$\text{Dry cows: } 0.00711 \times \text{BW} + 0.324 \times \text{CP} + 0.259$$

$$\times \text{NDF} + 8.05 \quad (\text{Equation 14-15b})$$

(Wilkerson et al., 1997)

$$\text{Lactating cows: } 2.63 \times \text{DMI} + 9.4 \quad (\text{Equation 14-15c})$$

(Nennich et al., 2005) where DMI is in kg/d, BW is in kg, and CP and NDF are percentages of diet DM.

REFERENCES

- Agle, M., A. N. Hristov, S. Zaman, C. Schneider, P. Ndegwa, and V. K. Vaddella. 2010. The effects of ruminally degraded protein on rumen fermentation and ammonia losses from manure in dairy cows. *J. Dairy Sci.* 93(4): 1625-1637.
- Alanis, P., M. Sorenson, M. Beene, C. Krauter, B. Shamp, and A. S. Hasson. 2008. Measurement of non-enteric emission fluxes of volatile fatty acids from a California dairy by solid phase micro-extraction with gas chromatography/mass spectrometry. *Atmospheric Environ.* 42(26):6417-6424.
- Alvarez-Fuentes, G., J. A. D. R. N. Appuhamy, and E. Kebreab. 2016. Prediction of phosphorus output in manure and milk by lactating dairy cows. *J. Dairy Sci.* 99(1):771-782.
- Appuhamy, J. A. D. R. N., A. B. Strathe, S. Jayasundara, C. Wagner-Riddle, J. Dijkstra, J. France, and E. Kebreab. 2013. Anti-methanogenic effects of monensin in dairy and beef cattle: A meta-analysis. *J. Dairy Sci.* 96(8):5161-5173.
- Appuhamy, J. A. D. R. N., L. E. Moraes, C. Wagner-Riddle, D. P. Casper, J. France, and E. Kebreab. 2014. Development of mathematical models to predict volume and nutrient composition of fresh manure from lactating Holstein cows. *Anim. Prod. Sci.* 54(11/12): 1927-1938.
- Appuhamy, J. A. D. R. N., J. France, and E. Kebreab. 2016. Models for predicting enteric methane emissions from dairy cows in North America, Europe, and Australia and New Zealand. *Global Change Biol.* 22(9):3039-3056.
- Appuhamy, J. A. D. R. N., L. E. Moraes, C. Wagner-Riddle, D. P. Casper, and E. Kebreab. 2018. Predicting manure volatile solid output of lactating dairy cows. *J. Dairy Sci.* 101(1):820-829.
- Bannink, A., H. Valk, and A. M. Van Vuuren. 1999. Intake and excretion of sodium, potassium, and nitrogen and the effects on urine production by lactating dairy cows. *J. Dairy Sci.* 82(5): 1008-1018.
- Bannink, A., J. France, S. Lopez, W. J. J. Gerrits, E. Kebreab, S. Tamminga, and J. Dijkstra. 2008. Modelling the implications of feeding strategy on rumen fermentation and functioning of the rumen wall. *Anim. Feed Sci. Technol.* 143(1-4):3-26.
- Bannink, A., M. W. van Schijndel, and J. Dijkstra. 2011. A model of enteric fermentation in dairy cows to estimate methane emission for the Dutch National Inventory Report using the IPCC tier 3 approach. *Anim. Feed Sci. Technol.* 166-167:603-618.
- Bemier, J. N., M. Undi, J. C. Plaizier, K. M. Wittenberg, G. R. Donohoe, and K. H. Ominski. 2012. Impact of prolonged cold exposure on dry matter intake and enteric methane emissions of beef cows overwintered on low-quality forage diets with and without supplemented wheat and corn dried distillers' grain with solubles. *Can. J. Anim. Sci.* 92(4):493-500.
- Bougouin, A., A. Leytem, J. Dijkstra, R. S. Dungan, and E. Kebreab. 2016. Nutritional and environmental effects on ammonia emissions from dairy cattle housing: A meta-analysis. *J. Environ. Qual.* 45(4): 1123-1132.
- Brock, E. H., Q. M. Ketterings, and M. McBride. 2006. Copper and zinc accumulation in poultry and dairy manure-amended fields. *Soil Sci.* 171(5):388-399.
- Burgos, S. A., J. G. Fadel, and E. J. Depeters. 2007. Prediction of ammonia emission from dairy cattle manure based on milk urea nitrogen: Relation of milk urea nitrogen to urine urea nitrogen excretion. *J. Dairy Sci.* 90(12):5499-508.
- Calsamiglia, S., A. Ferret, C. K. Reynolds, N. B. Kristensen, and A. M. van Vuuren. 2010. Strategies for optimizing nitrogen use by ruminants. *Animal* 4(7): 1184-1196.
- Capper, J. L. 2011. The environmental impact of beef production in the United States: 1977 compared with 2007. *J. Anim. Sci.* 89(12):4249-4261.
- Capper, J. L., R. A. Cady, and D. E. Bauman. 2009. The environmental impact of dairy production: 1944 compared with 2007. *J. Anim. Sci.* 87(6):2160-2167.
- Carter, W. P. L. 1994. Development of ozone reactivity scales for volatile organic compounds. *J. Air Waste Manage. Assoc.* 44(7):881-899.
- Castillo, A. R., E. Kebreab, D. E. Beever, and J. France. 2000. A review of efficiency of nitrogen utilisation in lactating dairy cows and its relationship with environmental pollution. *J. Anim. Feed Sci.* 9(1): 1-32.
- Castillo, A. R., E. Kebreab, D. E. Beever, J. H. Barbi, J. D. Sutton, H. C. Kirby, and J. France. 2001a. The effect of energy supplementation on nitrogen utilization in lactating dairy cows fed grass silage diets. *J. Anim. Sci.* 79(1):240-246.
- Castillo, A. R., E. Kebreab, D. E. Beever, J. H. Barbi, J. D. Sutton, H. C. Kirby, and J. France. 2001b. The effect of protein supplementation on nitrogen utilization in lactating dairy cows fed grass silage diets. *J. Anim. Sci.* 79(0):247-253.
- Castillo, A. R., N. R. St-Pierre, N. S. del Rio, and W. P. Weiss. 2013. Mineral concentrations in diets, water, and milk and their value in estimating on-farm excretion of manure minerals in lactating dairy cows. *J. Dairy Sci.* 96(5):3388-3398.
- Cavigelli, M. A., S. J. Del Grosso, M. A. Liebig, C. S. Snyder, P. E. Fixen, R. T. Venterea, A. B. Leytem, J. E. McLain, and D. B. Watts. 2012. US agricultural nitrous oxide emissions: Context, status, and trends. *Front. Ecol. Environ.* 10(10):537-546.
- Chung, M. Y., M. Beene, S. Ashkan, C. Krauter, and A. S. Hasson. 2010. Evaluation of non-enteric sources of non-methane volatile organic compound (NMVOC) emissions from dairies. *Atmospheric Environ.* 44(6):786-794.
- Clough, T. J., J. L. Ray, L. E. Buckthought, J. Calder, D. Baird, M. O'Callaghan, R. R. Sherlock, and L. M. Condron. 2009. The mitigation potential of hippuric acid on N₂O emissions from urine patches: An in situ determination of its effect. *Soil Biol. Biochem.* 41(10):2222-2229.
- Colmenero, J. J., and G. A. Broderick. 2006. Effect of dietary crude protein concentration on milk production and nitrogen utilization in lactating dairy cows. *J. Dairy Sci.* 89(5): 1704-1712.
- Dijkstra, J., O. Oenema, and A. Bannink. 2011. Dietary strategies to reducing N excretion from cattle: Implications for methane emissions. *Curr. Opin. Environ. Sustain.* 3(5):414-422.
- Dijkstra, J., J. France, J. L. Ellis, A. B. Strathe, E. Kebreab, and A. Bannink. 2013a. Production efficiency of ruminants: Feed, nitrogen and methane. Pp. 10-25 in *Sustainable Animal Agriculture*, E. Kebreab, ed. Wallingford, UK: CAB International.
- Dijkstra, J., O. Oenema, J. W. van Groenigen, J. W. Spek, A. M. van Vuuren, and A. Bannink. 2013b. Diet effects on urine composition of cattle and N₂O emissions. *Animal* 7:292-302.
- Dijkstra, J., C. K. Reynolds, E. Kebreab, A. Bannink, J. L. Ellis, J. France, and A. M. van Vuuren. 2013c. Challenges in ruminant nutrition: Towards minimal nitrogen losses in cattle. Pp. 47-58 in *Energy and Protein Metabolism and Nutrition in Sustainable Animal Production*, J. W. Oltjen, E. Kebreab, and H. Lapierre, eds. Wageningen, The Netherlands: Wageningen Academic.
- Dijkstra, J., A. Bannink, J. France, E. Kebreab, and S. van Gastelen. 2018. Short communication: Antimethanogenic effects of 3-nitrooxypropanol depend on supplementation dose, dietary fiber content, and cattle type. *J. Dairy Sci.* 101(10):9041-9047.
- Dou, Z., J. D. Ferguson, J. Fiorini, J. D. Toth, S. M. Alexander, L. E. Chase, C. M. Ryan, K. F. Knowlton, R. A. Kohn, A. B. Peterson, J. T. Sims, and Z. Wu. 2003. Phosphorus feeding levels and critical control points on dairy farms. *J. Dairy Sci.* 86(11):3787-3795.
- Eckard, R. J., C. Grainger, and C. A. M. de Klein. 2010. Options for the abatement of methane and nitrous oxide from ruminant production: A review. *Livestock Sci.* 130(1-3):47-56.
- Ellis, J. L., E. Kebreab, N. E. Odongo, B. W. McBride, E. K. Okine, and J. France. 2007. Prediction of methane production from dairy and beef cattle. *J. Dairy Sci.* 90(7):3456-3466.
- Ellis, J. L., A. Bannink, J. France, E. Kebreab, and J. Dijkstra. 2010. Evaluation of enteric methane prediction equations for dairy cows used in whole farm models. *Global Change Biol.* 16(12):3246-3256.
- Ellis, J. L., J. Dijkstra, J. France, A. J. Parsons, G. R. Edwards, S. Rasmussen, E. Kebreab, and A. Bannink. 2012. Effect of high-sugar grasses on methane emissions simulated using a dynamic model. *J. Dairy Sci.* 95(1):272-285.
- Eriksson, T., and B. O. Rustas. 2014. Effects on milk urea concentration, urine output, and drinking water intake from incremental doses of

- potassium bicarbonate fed to mid-lactation dairy cows. *J. Dairy Sci.* 97(7):4471-4484.
- Eugene, M., D. Masse, J. Chiquette, and C. Benchaar. 2008. Meta-analysis on the effects of lipid supplementation on methane production in lactating dairy cows. *Can. J. Anim. Sci.* 88(2):331-334.
- Ferket, P. R., E. van Heugten, T. A. T. G. van Kempen, and R. Angel. 2002. Nutritional strategies to reduce environmental emissions from nonruminants. *J. Anim. Sci.* 80(E. Suppl. 2):E168-E182.
- Filipy, J., B. Rumburg, G. Mount, H. Westberg, and B. Lamb. 2006. Identification and quantification of volatile organic compounds from a dairy. *Atmospheric Environ.* 40(8): 1480-1494.
- Frank, B., and C. Swensson. 2002. Relationship between content of crude protein in rations for dairy cows and milk yield, concentration of urea in milk and ammonia emissions. *J. Dairy Sci.* 85(7): 1829-1838.
- Frank, B., M. Persson, and G. Gustafsson. 2002. Feeding dairy cows for decreased ammonia emission. *Livestock Prod. Sci.* 76(1-2): 171-179.
- Fraser, M. D., H. R. Fleming, and J. M. Moorby. 2014. Traditional vs modern: Role of breed type in determining enteric methane emissions from cattle grazing as part of contrasting grassland-based systems. *PLoS ONE* 9(9):e107861.
- Glasser, F., A. Ferlay, and Y. Chilliard. 2008. Oilseed lipid supplements and fatty acid composition of cow milk: A meta-analysis. *J. Dairy Sci.* 91(12):4687-4703.
- Goel, G., H. P. Makkar, and K. Becker. 2009. Inhibition of methanogens by bromochloromethane: Effects on microbial communities and rumen fermentation using batch and continuous fermentations. *Br. J. Nutr.* 101:1484-1492.
- Grainger, C., and K. A. Beauchemin. 2011. Can enteric methane emissions from ruminants be lowered without lowering their production? *Anim. Feed Sci. Technol.* 166-167:308-320.
- Groenigen, J. W. v., P. J. Kuikman, W. J. M. d. Groot, and G. L. Velthof. 2005. Nitrous oxide emission from urine-treated soil as influenced by urine composition and soil physical conditions. *Soil Biol. Biochem.* 37(3):463-473.
- Guyader, J., M. Eugene, M. Doreau, D. P. Morgavi, C. Gerard, and C. Martin. 2017. Tea saponin reduced methanogenesis in vitro but increased methane yield in lactating dairy cows. *J. Dairy Sci.* 100(3): 1845-1855.
- Haan, M. M., J. R. Russell, W. J. Powers, J. L. Kovar, and J. L. Benning. 2006. Grazing management effects on sediment and phosphorus in surface runoff. *Rangeland Ecol. Manage.* 59(6):607-615.
- Hafner, S. D., F. Montes, C. A. Rotz, and F. Mitloehner. 2010. Ethanol emission from loose cow silage and exposed silage particles. *Atmospheric Environ.* 44(34):4172-4180.
- Haque, M. N. H., R. L. Rulquin, A. Andrade, P. Faverdin, J. L. Peyraud, and S. Lemosquet. 2012. Milk protein synthesis in response to the provision of an "ideal" amino acid profile at 2 levels of metabolizable protein supply in dairy cows. *J. Dairy Sci.* 95(10):5876-5887.
- Harter, T., H. Davis, M. C. Mathews, and R. D. Meyer. 2002. Shallow groundwater quality on dairy farms with irrigated forage crops. *J. Contam. Hydrol.* 55(3-4):287-315.
- Hill, S. R., K. F. Knowlton, E. Kebreab, J. France, and M. D. Hanigan. 2008. A model of phosphorus digestion and metabolism in the lactating dairy cow. *J. Dairy Sci.* 91:2021-2032.
- Honan, M., X. Feng, J. Tricarico, and E. Kebreab. 2021. Feed additives as a strategic approach to reduce enteric methane production in cattle: Modes of action, effectiveness, and safety. *Anim. Prod. Sci.* <https://www.publish.csiro.au/paper/AN20295>. Accessed July 13, 2021.
- Howard, C. J., A. Kumar, I. Malkina, F. Mitloehner, P. G. Green, R. G. Flocchini, and M. J. Kleeman. 2010. Reactive organic gas emissions from livestock feed contribute significantly to ozone production in central California. *Environ. Sci. Technol.* 44(7):2309-2314.
- Hristov, A. N., W. Hazen, and J. W. Ellsworth. 2007. Efficiency of use of imported magnesium, sulfur, copper, and zinc on Idaho dairy farms. *J. Dairy Sci.* 90(6):3034-3043.
- Hristov, A. N., M. Hanigan, A. Cole, R. Todd, T. A. McAllister, P. M. Ndegwa, and A. Rotz. 2011. Review: Ammonia emissions from dairy farms and beef feedlots. *Can. J. Anim. Sci.* 91(1): 1-35.
- Hristov, A. N., J. Oh, J. L. Firkins, J. Dijkstra, E. Kebreab, G. Waghom, H. P. S. Makkar, A. T. Adesogan, W. Yang, C. Lee, P. J. Gerber, B. Henderson, and J. M. Tricarico. 2013a. Special topics—Mitigation of methane and nitrous oxide emissions from animal operations: I. A review of enteric methane mitigation options. *J. Anim. Sci.* 91(11):5045-5069.
- Hristov, A. N., T. Ott, J. Tricarico, A. Rotz, G. Waghom, A. Adesogan, J. Dijkstra, F. Montes, J. Oh, E. Kebreab, S. J. Oosting, P. J. Gerber, B. Henderson, H. P. S. Makkar, and J. L. Firkins. 2013b. Special topics—Mitigation of methane and nitrous oxide emissions from animal operations: III. A review of animal management mitigation options. *J. Anim. Sci.* 91(11):5095-5113.
- Hristov, A. N., J. Oh, F. Giallongo, T. W. Frederick, M. T. Harper, H. L. Weeks, A. F. Branco, P. J. Moate, M. H. Deighton, S. R. O. Williams, M. Kindermann, and S. Duval. 2015. An inhibitor persistently decreased enteric methane emission from dairy cows with no negative effect on milk production. *Proc. Natl. Acad. Sci. U. S. A.* 112(34): 10663-10668.
- Huhtanen, P., and A. N. Hristov. 2009. A meta-analysis of the effects of dietary protein concentration and degradability on milk protein yield and milk N efficiency in dairy cows. *J. Dairy Sci.* 92(7):3222-3232.
- Huhtanen, P., J. I. Nousiainen, M. Rinne, K. Kytola, and H. Khalili. 2008. Utilization and partition of dietary nitrogen in dairy cows fed grass silage-based diets. *J. Dairy Sci.* 91(9):3589-3599.
- IPCC (Intergovernmental Panel on Climate Change). 1997. Revised 1996 IPCC Guidelines for National Greenhouse Gas Inventories: Reference Manual (Volume 3). Kanagawa, Japan: Task Force on National Greenhouse Gas Inventories.
- IPCC. 2006. Guidelines for National Greenhouse Gas Inventories. Volume 4: Agriculture, Forestry and Other Land Use. Prepared by the National Greenhouse Gas Inventories Program. Kanagawa, Japan: Institute for Global Environmental Strategies (IGES).
- IPCC. 2013. Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge, UK: Cambridge University Press.
- IPCC. 2019. 2019 refinement to the 2006 IPCC guidelines for National Greenhouse Gas Inventories. In Emissions from Livestock and Manure Management, E. Calvo Buendia, K. Tanabe, A. Kranjc, J. Baasansuren, M. Fukuda, S. Ngarize, A. Osako, Y. Pyrozhenko, P. Shermanau, and S. Federici, eds. https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch10_Livestock.pdf. Accessed September 21, 2020.
- Jayasundara, S., J. A. D. R. N. Appuhamy, E. Kebreab, and C. Wagner-Riddle. 2016. Methane and nitrous oxide emissions from Canadian dairy farms and mitigation options: An updated review. *Can. J. Anim. Sci.* 96(3):306-331.
- Johnson, A. C. B., K. F. Reed, and E. Kebreab. 2016. Short communication: Evaluation of nitrogen excretion equations from cattle. *J. Dairy Sci.* 99(9):7669-7678.
- Josan, M. S., V. D. Nair, W. G. Harris, and D. Herrera. 2005. Associated release of magnesium and phosphorus from active and abandoned dairy soils. *J. Environ. Qual.* 34(1): 184-191.
- Kampa, M., and E. Castanas. 2008. Human health effects of air pollution. *Environ. Pollut.* 151(2):362-367.
- Kebreab, E. 2013. Sustainable Animal Agriculture. Boston, MA: CABI.
- Kebreab, E., J. France, D. E. Beever, and A. R. Castillo. 2001. Nitrogen pollution by dairy cows and its mitigation by dietary manipulation. *Nutrient Cycling Agroecosyst.* 60(1-3):275-285.
- Kebreab, E., J. France, J. A. N. Mills, R. Allison, and J. Dijkstra. 2002. A dynamic model of N metabolism in the lactating dairy cow and an assessment of impact of N excretion on the environment. *J. Anim. Sci.* 80(1):248-259.

- Kebreab, E., J. France, J. D. Sutton, L. A. Crompton, and D. E. Beever. 2005a. Effect of energy and protein supplementation on phosphorus utilization in lactating dairy cows. *J. Anim. Feed Sci.* 14(1):63–77.
- Kebreab, E., M. A. Shah, D. E. Beever, D. J. Humphries, J. D. Sutton, J. France, and J. Mueller-Harvey. 2005b. Effects of contrasting forage diets on phosphorus utilisation in lactating dairy cows. *Livestock Prod. Sci.* 93(2): 125–135.
- Kebreab, E., K. A. Johnson, S. L. Archibeque, D. Pape, and T. Wirth. 2008a. Model for estimating enteric methane emissions from United States dairy and feedlot cattle. *J. Anim. Sci.* 86(10):2738–2748.
- Kebreab, E., N. E. Odongo, B. W. McBride, M. D. Hanigan, and J. France. 2008b. Phosphorus utilization and environmental and economic implications of reducing phosphorus pollution from Ontario dairy cows. *J. Dairy Sci.* 91(1):241–246.
- Kebreab, E., A. B. Strathe, J. Dijkstra, J. A. N. Mills, C. K. Reynolds, L. A. Crompton, T. Yan, and J. France. 2010. Energy and Protein Interactions and Their Effects on Nitrogen Excretion in Dairy Cows. Pp. 417–425 in 3rd EAAP International Symposium on Energy and Protein Metabolism and Nutrition, Parma, Italy, 6–10 September, 2010. G. M. Crovetto, ed. Wageningen, The Netherlands: Wageningen Academic Publishers.
- Kebreab, E., A. V. Hansen, and A. B. Strathe. 2012. Animal production for efficient phosphate utilization: From optimized feed to high efficiency livestock. *Curr. Opin. Biotechnol.* 23(6):872–877.
- Kebreab, E., A. V. Hansen, and A. B. Leytem. 2013. Feed management practices to reduce manure phosphorus excretion in dairy cattle. *Adv. Anim. Biosci.* 4(1):37–41.
- Kellogg, R. L., C. H. Lander, Moffitt D., and N. Gollehon. 2000. Manure nutrients relative to the capacity of cropland and pastureland to assimilate nutrients: Spatial and temporal trends for the United States. U.S. Department of Agriculture Publication No. nps00-0579. https://www.nrcs.usda.gov/Internet/FSE_DOCUMENTS/nrcs_143_012133.pdf. Accessed November 1, 2018.
- Kinley, R. D., R. de Nys, M. J. Vucko, L. Machado, and N. W. Tomkins. 2016. The red macroalgae *Asparagopsis taxiformis* is a potent natural antimethanogenic that reduces methane production during in vitro fermentation with rumen fluid. *Anim. Prod. Sci.* 56(2–3):282–289.
- Klop, G., J. L. Ellis, A. Bannink, E. Kebreab, J. France, and J. Dijkstra. 2013. Meta-analysis of factors that affect the utilization efficiency of phosphorus in lactating dairy cows. *J. Dairy Sci.* 96(6):3936–3949.
- Knapp, J. R., G. L. Laur, P. A. Vadas, W. P. Weiss, and J. M. Tricarico. 2014. Invited review: Enteric methane in dairy cattle production: Quantifying the opportunities and impact of reducing emissions. *J. Dairy Sci.* 97(6):3231–3261.
- Lapierre, H., R. Berthiaume, G. Raggio, M. C. Thivierge, L. Doepel, D. Pacheco, P. Dubreuil, and G. E. Lobley. 2005. The route of absorbed nitrogen into milk protein. *Anim. Sci.* 80:11–22.
- Law, R. A., F. J. Young, D. C. Patterson, D. J. Kilpatrick, A. R. Wylie, and C. S. Mayne. 2009. Effect of dietary protein content on animal production and blood metabolites of dairy cows during lactation. *J. Dairy Sci.* 92(3): 1001–1012.
- Lee, C., A. N. Hristov, T. Cassidy, and K. Heyler. 2011. Nitrogen isotope fractionation and origin of ammonia nitrogen volatilized from cattle manure in simulated storage. *Atmosphere* 2(3):256–270.
- Leytem, A. B., R. S. Dungan, D. L. Bjomeberg, and A. C. Koehn. 2011. Emissions of ammonia, methane, carbon dioxide, and nitrous oxide from dairy cattle housing and manure management systems. *J. Environ. Qual.* 40(5): 1383–1394.
- Leytem, A. B., R. S. Dungan, D. L. Bjomeberg, and A. C. Koehn. 2013. Greenhouse gas and ammonia emissions from an open-freestall dairy in southern Idaho. *J. Environ. Qual.* 42(1):10–20.
- Li, C. S., W. Salas, R. H. Zhang, C. Krauter, A. Rotz, and F. Mitloehner. 2012. Manure-DNDC: A biogeochemical process model for quantifying greenhouse gas and ammonia emissions from livestock manure systems. *Nutrient Cycling Agroecosyst.* 93(2): 163–200.
- Luo, J., C. A. M. d. Klein, S. F. Ledgard, and S. Saggar. 2010. Management options to reduce nitrous oxide emissions from intensively grazed pastures: A review. *Agric. Ecosyst. Environ.* 136(3/4):282–291.
- Martens, W., M. Martinec, R. Zapirain, M. Stark, E. Hartung, and U. Palmgren. 2001. Reduction potential of microbial, odour and ammonia emissions from a pig facility by biofilters. *Int. J. Hyg. Environ. Health* 203(4):335–345.
- Martin, C., D. P. Morgavi, and M. Doreau. 2010. Methane mitigation in ruminants: From microbe to the farm scale. *Animal* 4(3):351–365.
- McQuilling, A. M., and P. J. Adams. 2015. Semi-empirical process-based models for ammonia emissions from beef, swine, and poultry operations in the United States. *Atmospheric Environ.* 120:127–136.
- Meals, D. W., S. A. Dressing, and T. E. Davenport. 2010. Lag time in water quality response to best management practices: A review. *J. Environ. Qual.* 39(1):85–96.
- Meyer, D., and P. H. Robinson. 2007. Dairy Nutritionists' Roles in Nutrient Use: Recommendations for Feed Nutrient Records Analyses. University of California Agriculture and Natural Resources Publication 8278. <https://anrcatalog.ucanr.edu/pdf/8278.pdf>. Accessed November 1, 2018.
- Mills, J. A. N., J. Dijkstra, A. Bannink, S. B. Cammell, E. Kebreab, and J. France. 2001. A mechanistic model of whole-tract digestion and methanogenesis in the lactating dairy cow: Model development, evaluation, and application. *J. Anim. Sci.* 79(6): 1584–1597.
- Mills, J. A. N. E. Kebreab, C. M. Yates, L. A. Crompton, S. B. Cammell, M. S. Dhanoa, R. E. Agnew, and J. France. 2003. Alternative approaches to predicting methane emissions from dairy cows. *J. Anim. Sci.* 81(12):3141–3150.
- Moe, P. W., and H. F. Tyrrell. 1979. Methane production in dairy-cows. *J. Dairy Sci.* 62(10): 1583–1586.
- Montes, F., R. Meinen, C. Dell, A. Rotz, A. N. Hristov, J. Oh, G. Waghom, P. J. Gerber, B. Henderson, H. P. Makkar, and J. Dijkstra. 2013. Special topics—Mitigation of methane and nitrous oxide emissions from animal operations: II. A review of manure management mitigation options. *J. Anim. Sci.* 91(11):5070–5094.
- Moraes, L. E., J. E. Wilen, P. H. Robinson, and J. G. Fadel. 2012. A linear programming model to optimize diets in environmental policy scenarios. *J. Dairy Sci.* 95(3): 1267–1282.
- Moraes, L. E., A. B. Strathe, J. G. Fadel, D. P. Casper, and E. Kebreab. 2014. Prediction of enteric methane emissions from cattle. *Global Change Biol.* 20(7):2140–2148.
- Muck, R. E. 1988. Factors influencing silage quality and their implications for management. *J. Dairy Sci.* 71(11):2992–3002.
- Munoz, C., T. Yan, D. A. Wills, S. Murray, and A. W. Gordon. 2012. Comparison of the sulfur hexafluoride tracer and respiration chamber techniques for estimating methane emissions and correction for rectum methane output from dairy cows. *J. Dairy Sci.* 95(6):3139–3148.
- Murphy, M. R., R. L. Baldwin, and L. J. Koong. 1982. Estimation of stoichiometric parameters for rumen fermentation of roughage and concentrate diets. *J. Anim. Sci.* 55(2):411–421.
- Murray, R. M., A. M. Bryant, and R. A. Leng. 1976. Rates of production of methane in rumen and large-intestine of sheep. *Br. J. Nutr.* 36(1): 1–14.
- Nennich, T. D., J. H. Harrison, L. M. Van Wieringen, D. Meyer, A. J. Heinrichs, W. P. Weiss, N. R. St-Pierre, R. L. Kincaid, D. L. Davidson, and E. Block. 2005. Prediction of manure and nutrient excretion from dairy cattle. *J. Dairy Sci.* 88(10):3721–3733.
- Nielsen, N. I., H. Volden, M. Akerlind, M. Brask, A. L. F. Hellwing, T. Storlien, and J. Bertilsson. 2013. A prediction equation for enteric methane emission from dairy cows for use in NorFor. *Acta Agric. Scand. A Anim. Sci.* 63(3): 126–130.
- Nousiainen, J., M. Rinne, and P. Huhtanen. 2009. A meta-analysis of feed digestion in dairy cows: 1. The effects of forage and concentrate factors on total diet digestibility. *J. Dairy Sci.* 92(10):5019–5030.
- NRC (National Research Council). 2001. Nutrient Requirements of Dairy Cattle. 7th rev. ed. Washington, DC: National Academy Press.

- NRC. 2003. Air Emissions from Animal Feed Intake and Operations. Washington, DC: The National Academies Press.
- Oenema, O., A. Bannink, S. G. Sommer, J. W. Van Groenigen, and G. L. Velthof. 2008. Gaseous nitrogen emissions from livestock farming systems. Pp. 395-441 in *Nitrogen in the Environment: Sources, Problems, and Management*. 2nd ed. London, UK: Elsevier.
- Pagans, E., X. Font, and A. Sanchez. 2007. Coupling composting and bio-filtration for ammonia and volatile organic compound removal. *Biosyst. Eng.* 97(4):491-500.
- Pasten-Zapata, E., R. Ledesma-Ruiz, T. Harter, A. I. Ramirez, and J. Mahlknecht. 2014. Assessment of sources and fate of nitrate in shallow groundwater of an agricultural area by using a multi-tracer approach. *Sci. Total Environ.* 470:855-864.
- Petersen, S. O., and S.G. Sommer. 2011. Ammonia and nitrous oxide interactions: Roles of manure organic matter management. *Anim. Feed Sci. Technol.* 166:503-513.
- Pierzynski, G. M., J. T. Sims, and G. F. Vance. 1994. *Soils and Environmental Quality*. Boca Raton, FL: Lewis.
- Place, S. E., and F. M. Mitloehner. 2013. Air quality issues in sustainability: Greenhouse gases and volatile organic compounds. Pp. 124-136 in *Sustainable Animal Agriculture*, E. Kebreab, ed. Oxfordshire, UK: CABI Publishing.
- Powell, J. M., and G. A. Broderick. 2011. Transdisciplinary soil science research: Impacts of dairy nutrition on manure chemistry and the environment. *Soil Sci. Soc. Am. J.* 75(6):2071-2078.
- Powell, J. M., M. A. Wattiaux, and G. A. Broderick. 2011. Short communication: Evaluation of milk urea nitrogen as a management tool to reduce ammonia emissions from dairy farms. *J. Dairy Sci.* 94(9):4690-4694.
- Powell, J. M., C. A. Rotz, and M. A. Wattiaux. 2014. Potential use of milk urea nitrogen to abate atmospheric nitrogen emissions from Wisconsin dairy farms. *J. Environ. Qual.* 43(4): 1169—1175.
- Reed, K. F., L. E. Moraes, J. G. Fadel, D. P. Casper, J. Dijkstra, J. France, and E. Kebreab. 2014. Prediction of nitrogen use in dairy cattle: A multivariate Bayesian approach. *Anim. Prod. Sci.* 54:1918-1926.
- Reed, K. F., L. E. Moraes, D. P. Casper, and E. Kebreab. 2015. Predicting nitrogen excretion from cattle. *J. Dairy Sci.* 98(5):3025—3035.
- Reynal, S. M., and G. A. Broderick. 2005. Effect of dietary level of rumen-degraded protein on production and nitrogen metabolism in lactating dairy cows. *J. Dairy Sci.* 88(11):4045-4064.
- Reynolds, C. K., and N. B. Kristensen. 2008. Nitrogen recycling through the gut and the nitrogen economy of ruminants: An asynchronous symbiosis. *J. Anim. Sci.* 86(14, Suppl.):E293-E305.
- Roque, B. M., J. K. Salwen, R. Kinley, and E. Kebreab. 2019. Inclusion of *Asparagopsis armata* in lactating dairy cows' diet reduces enteric methane emission by over 50 percent. *J. Clean. Prod.* 234:132-138.
- Rosenstock, T. S., D. Liptzin, K. Dzurella, A. Fryjoff-Hung, A. Hollander, V. Jensen, A. King, G. Kourakos, A. McNally, G. S. Pettygrove, J. Quinn, J. H. Viers, T. P. Tomich, and T. Harter. 2014. Agriculture's contribution to nitrate contamination of Californian groundwater (1945-2005). *J. Environ. Qual.* 43(3):895-907.
- Sauvant, D., M. Eugene, S. Giger-Reverdin, H. Archimede, and M. Doreau. 2014. Relationship between CH₄ and urinary N outputs in ruminants fed forages: A meta-analysis of the literature. *Anim. Prod. Sci.* 54(9): 1423-1427.
- Shaw, S. L., F. M. Mitloehner, W. Jackson, E. J. Depeters, J. G. Fadel, P. H. Robinson, R. Holzinger, and A. H. Goldstein. 2007. Volatile organic compound emissions from dairy cows and their waste as measured by proton-transfer-reaction mass spectrometry. *Environ. Sci. Technol.* 41(4): 1310—1316.
- Smith, R. A., and R. B. Alexander. 2000. Sources of nutrients in the nation's watersheds. Pp. 13-21 in *Proceedings of the Managing Nutrients and Pathogens from Animal Agriculture*. Camp Hill, PA: Natural Resource, Agriculture, and Engineering Service.
- Spek, J. W., A. Bannink, G. Gort, W. H. Hendriks, and J. Dijkstra. 2012. Effect of sodium chloride intake on urine volume, urinary urea excretion, and milk urea concentration in lactating dairy cattle. *J. Dairy Sci.* 95(12):7288—7298.
- Spek, J. W., A. Bannink, G. Gort, W. H. Hendriks, and J. Dijkstra. 2013a. Interaction between dietary content of protein and sodium chloride on milk urea concentration, urinary urea excretion, renal recycling of urea, and urea transfer to the gastrointestinal tract in dairy cows. *J. Dairy Sci.* 96(9):5734-5745.
- Spek, J. W., J. Dijkstra, G. Van Duinkerken, and A. Bannink. 2013b. A review of factors influencing milk urea concentration and its relationship with urinary urea excretion in lactating dairy cattle. *J. Agric. Sci.* 151(3):407-423.
- Spek, J. W., J. Dijkstra, G. van Duinkerken, W. H. Hendriks, and A. Bannink. 2013c. Prediction of urinary nitrogen and urinary urea nitrogen excretion by lactating dairy cattle in northwestern Europe and North America: A meta-analysis. *J. Dairy Sci.* 96(7):4310-4322.
- Tamminga, S. 1992. Nutrition management of dairy-cows as a contribution to pollution-control. *J. Dairy Sci.* 75(1):345—357.
- Taylor, K., J. A. D. R. N. Appuhamy, J. Dijkstra, and E. Kebreab. 2018. Development of mathematical models to predict calcium, magnesium and selenium excretion from lactating Holstein cows. *Anim. Prod. Sci.* 58(3):489-198.
- Tewoldebrihan, T. A., J. A. D. R. N. Appuhamy, J. J. Lee, M. Niu, S. Seo, S. Jeong, and E. Kebreab. 2017. Exogenous beta-mannanase improves feed conversion efficiency and reduces somatic cell count in dairy cattle. *J. Dairy Sci.* 100(1):244-252.
- Thoma, G., J. Popp, D. Nutter, D. Shonnard, R. Ulrich, M. Matlock, D. S. Kim, Z. Neiderman, N. Kemper, C. East, and F. Adorn. 2013. Greenhouse gas emissions from milk production and consumption in the United States: A cradle-to-grave life cycle assessment circa 2008. *Int. Dairy J.* 31: S3-S 14.
- Townsend, A. R., R. W. Howarth, F. A. Bazzaz, M. S. Booth, C. C. Cleveland, S. K. Collinge, A. P. Dobson, P. R. Epstein, D. R. Keeney, M. A. Mallin, C. A. Rogers, P. Wayne, and A. H. Wolfe. 2003. Human health effects of a changing global nitrogen cycle. *Front. Ecol. Environ.* 1(5):240-246.
- USDA, NASS (U.S. Department of Agriculture, National Agricultural Statistics Service). 2005. Farms, Land in Farms, and Livestock Operations: 2004 Summary, <https://downloads.usda.library.cornell.edu/usda-esmis/files/5712m6524/47429c799/05741v28n/FarmLandIn-01-31-2005.pdf>.
- USDA, NASS. 2013. Farms, Land in Farms, and Livestock Operations: 2012 Summary, <https://downloads.usda.library.cornell.edu/usda-esmis/files/5712m6524/3t945t54c/m3014150/FarmLandIn-02-17-2012.pdf>.
- USEPA (U.S. Environmental Protection Agency). 2009. The next generation of tools and actions to restore water quality in the Chesapeake Bay. A revised report fulfilling section 202a of executive order 13508. Washington, DC: USEPA.
- USEPA. 2011. Volatile Organic Compounds (VOCs): Technical Overview. Washington, DC: USEPA.
- USEPA. 2014. Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2012. Washington, DC: USEPA.
- Valk, H., L. B. J. Sebek, and A. C. Beynen. 2002. Influence of phosphorus intake on excretion and blood plasma and saliva concentrations of phosphorus in dairy cows. *J. Dairy Sci.* 85(10):2642-2649.
- van der Stelt, B., P. C. J. van Vliet, J. W. Reijs, E. J. M. Temminghoff, and W. H. van Riemsdijk. 2008. Effects of dietary protein and energy levels on cow manure excretion and ammonia volatilization. *J. Dairy Sci.* 91(12):4811—4821.
- van Groenigen, J. W., R. L. M. Schils, G. L. Velthof, P. J. Kuikman, D. A. Oudendag, and O. Oenema. 2008. Mitigation strategies for greenhouse gas emissions from animal production systems: Synergy between measuring and modelling at different scales. *Aust. J. Exp. Agric.* 48(1-2):46-53.
- van Middelaar, C. E., P. B. M. Berentsen, J. Dijkstra, and I. J. M. De Boer. 2013. Evaluation of a feeding strategy to reduce greenhouse gas emissions from dairy farming: The level of analysis matters. *Agric. Syst.* 121:9-22.
- van Middelaar, C. E., J. Dijkstra, P. B. M. Berentsen, and I. J. M. De Boer. 2014. Cost-effectiveness of feeding strategies to reduce greenhouse gas emissions from dairy farming. *J. Dairy Sci.* 97(4):2427-2439.

- van Zijderveld, S. M., W. J. J. Gerrits, J. A. Apajalahti, J. R. Newbold, J. Dijkstra, R. A. Leng, and H. B. Perdok. 2010. Nitrate and sulfate: Effective alternative hydrogen sinks for mitigation of ruminal methane production in sheep. *J. Dairy Sci.* 93(12):5856—5866. van Zijderveld, S. M., W. J. J. Gerrits, J. Dijkstra, J. R. Newbold, R. B. A. Hulshof, and H. B. Perdok. 2011. Persistency of methane mitigation by dietary nitrate supplementation in dairy cows. *J. Dairy Sci.* 94(8):4028-1038.
- Vitti, D. M. S. S., and E. Kebreab. 2010. *Phosphorus and Calcium Utilization and Requirements in Farm Animals*. Wallingford, UK: CABI Publishing, von Keyserlingk, M. A. G., N. P. Martin, E. Kebreab, K. F. Knowlton, R. J. Grant, M. Stephenson, C. J. Sniffen, J. R. Hamer, A. D. Wright, and S. I. Smith. 2013. Invited review: Sustainability of the US dairy industry. *J. Dairy Sci.* 96(91):5405-5425.
- Wattiaux, M. A., and K. L. Karg. 2004. Protein level for alfalfa and com silage-based diets: II. Nitrogen balance and manure characteristics. *J. Dairy Sci.* 87(11):3492-3502.
- Weiss, W. P., and D. J. Wyatt. 2004. Macromineral digestion by lactating dairy cows: Estimating phosphorus excretion via manure. *J. Dairy Sci.* 87(7):2158-2166.
- Weiss, W. P., L. B. Willett, N. R. St-Pierre, D. C. Borger, T. R. McKelvey, and D. J. Wyatt. 2009. Varying forage type, metabolizable protein concentration, and carbohydrate source affects manure excretion.

- manure ammonia, and nitrogen metabolism of dairy cows. *J. Dairy Sci.* 92(11):5607—5619.
- White, R. R., and J. L. Capper. 2014. Precision diet formulation to improve performance and profitability across various climates: Modeling the implications of increasing the formulation frequency of dairy cattle diets. *J. Dairy Sci.* 97(3): 1563-1577.
- Wick, K., C. Heumesser, and E. Schmid. 2012. Groundwater nitrate contamination: Factors and indicators. *J. Environ. Manage.* 111:178-186.
- Wilkerson, V. A., D. R. Mertens, and D. P. Casper. 1997. Prediction of excretion of manure and nitrogen by Holstein dairy cattle. *J. Dairy Sci.* 80(12):3193-3204.
- Winsten, J. R., C. D. Kerchner, A. Richardson, A. Lichau, and J. M. Hyman. 2010. Trends in the Northeast dairy industry: Large-scale modern confinement feeding and management-intensive grazing. *J. Dairy Sci.* 93(4): 1759—1769.
- Wu, Z., L. D. Satter, and R. Sojo. 2000. Milk production, reproductive performance, and fecal excretion of phosphorus by dairy cows fed three amounts of phosphorus. *J. Dairy Sci.* 83(5): 1028—1041.
- Wu, Z., L. D. Satter, A. J. Blohowiak, R. H. Stauffacher, and J. H. Wilson. 2001. Milk production, estimated phosphorus excretion, and bone characteristics of dairy cows fed different amounts of phosphorus for two or three years. *J. Dairy Sci.* 84(7): 1738—1748.